EnergyLens

Energy intelligence, EU-compliance reporting and decision-support for European buildings, on Microsoft Fabric — EXIST-Gründungsstipendium application

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1. Idea sketch (the 60-second read)

The problem. Buildings consume about **40 % of EU energy** and produce about **36 % of energy-related CO_2** — the single largest sector. Yet most building portfolios are still run on monthly utility bills and spreadsheets. They cannot see the anomalies that waste energy nightly, cannot keep up with a fast-tightening wall of EU regulation (EPBD minimum standards, EnEfG, GEG, CSRD/ESRS, the CO_2 cost-allocation law, the EU Battery Regulation), and cannot prioritise the cheapest path to decarbonisation. Large vendors (Schneider, Siemens, Measurabl) serve the top of the market; the **mid-market and residential property managers are underserved**.

The solution. EnergyLens is a **Microsoft Fabric-native** energy-intelligence, EU-compliance and decision-support platform. It unifies meter, sensor, weather, solar and battery data into one auditable data model and delivers it through five layers: a **10-page analytics report**, a **multi-tenant web application**, an **AI copilot** that answers questions by running real queries over the data (tool use, not text summaries) and can take action, a **compliance & ESG reporting engine** that produces the actual regulatory documents a building owner needs, and an emerging **execution marketplace** that closes the loop from insight to action to delivery.

The innovation. EnergyLens is not a dashboard. Its defensible core is (a) an **agentic AI copilot** that reasons over a building-energy Lakehouse through typed tools, (b) a **regulator-grade, fully traceable carbon-lineage methodology** for ESRS/VSME reporting, and (c) **multi-protocol IoT fault detection** (ASHRAE Guideline 36) delivered at mid-market prices. The funded year develops these from a working prototype into a defensible, validated, research-anchored product (see §4).

The market. The Building Energy Management Systems (BEMS) software market is independently estimated at roughly **US $8.4 bn in 2026 → ~$24 bn by 2035 (~8.7 % CAGR)**. The focus is the **DACH mid-market** — the highest-value EU compliance market and the founder's home market — with a second vertical in **residential property management (Hausverwaltung)**.

Why this team. The founder combines genuine **energy-engineering domain knowledge** (B.Sc. Energy Engineering, M.Sc. Energy Management) with modern **data-engineering skill** (Microsoft DP-600) and has already built the full platform — pipeline, analytics, web app, AI copilot and a 13-report compliance engine — **solo, end-to-end**. This domain × data-engineering intersection is rare.

The 12-month goal. Convert a working prototype + a first pilot into an **incorporated company (UG)** with **5+ paying customers** and **€5–10k MRR**, a **published carbon-methodology paper**, and a defensible agentic-AI product — a fundable, post-grant business.

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2. The problem & why now

European real estate sits at the centre of the EU's climate agenda, and the regulatory pressure is now **binding, not aspirational**:

- **EPBD recast (Directive (EU) 2024/1275):** national minimum-energy-performance standards for non-residential buildings by **1 Jan 2027**; worst-performing **16 % improved by 2030, 26 % by 2033**; new buildings zero-emission from 2030.
- **EnEfG (Germany):** organisations above ~**2.5–7.5 GWh/year** must run energy-management and an economic-measures plan; larger users need ISO 50001 / DIN EN 16247 audits.
- **GEG §71 (Germany):** new heating systems must use **≥ 65 % renewable energy**; minimum envelope U-values for renovations.
- **CO_2 cost-allocation law (CO2KostAufG):** every landlord must split the carbon price on heating fuel with tenants on a 10-step model — an **annual, mandatory** calculation.
- **CSRD / ESRS E1:** sustainability disclosure, narrowed by the 2026 Omnibus to firms **> 1,000 employees & > €450 m turnover**, with smaller firms reporting voluntarily via **VSME**.
- **EU Battery Regulation (2023/1542):** carbon-footprint and recycled-content disclosure; digital battery passport from 2027.

At the same time the structural reasons to act all arrived together: the post-2022 energy-price shock made energy a board-level topic; **Microsoft Fabric** reached general availability (2024); IoT sensor costs collapsed; and AI made pattern recognition affordable. Owners face rising compliance exposure with no modern, affordable tooling — a real, time-bound demand.

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3. The solution & what is already built

The product is **not a concept — it already runs end-to-end**. That de-risks the venture and is itself evidence of execution capability. Built solo, April–June 2026:

3.1 Data foundation

A full Microsoft Fabric **medallion architecture** (Bronze → Silver → Gold) on OneLake, ~**57 Lakehouse tables**, with a **single sourced reference layer** for emission factors, tariffs and envelope U-values by construction vintage. **Real external data** is already integrated: Open-Meteo weather (real HDD/CDD), ENTSO-E day-ahead prices, ElectricityMaps grid-carbon intensity.

3.2 Analytics — 10-page Power BI report

Consumption, building deep-dive, anomalies, forecasting, decision-support, sustainability/GHG, HVAC, real-time IoT, battery ROI and **solar (on-site generation & self-consumption)** — on a custom DAX model (50+ measures, building-type-aware thresholds), reading live via **DirectLake**.

3.3 Multi-tenant web application

Next.js + FastAPI + Azure PostgreSQL, with **three-provider authentication** (Microsoft Entra ID, Google, email) and a **three-layer access model** (row-level security on data, module gating per subscription, commercial tiers). Custom pages — portfolio, building, solar, actions, alerts — read Fabric through the SQL Analytics Endpoint. Includes **Stripe billing**, an **audit log**, **PDF export**, an in-app **glossary/tooltip** layer, an **alerts triage** queue, accessibility and end-to-end tests, plus public **/demo** and **/pricing**.

3.4 AI copilot

Six production tools over the Lakehouse (query KPIs, compare buildings, list recommendations, get anomalies, simulate battery scenarios, update action status), server-sent-events streaming, and a **provider abstraction** (Anthropic / Azure OpenAI / Mock). Most BI "copilots" summarise a page; this one **runs real queries and can write back** — it can act, not only describe.

3.5 Compliance & ESG reporting engine

A generic report engine with a template registry produces the **actual documents** an owner needs — **13 reports built**, each framed honestly as *reporting support, not legal or audited filing*:

- **German-mandatory set:** CO_2 cost-allocation (CO2KostAufG, 10-step Stufenmodell), GEG conformity, Energieausweis pre-assessment.
- **Voluntary / high-demand:** GHG Inventory (GHG Protocol, dual-scope), **ESRS E-1** (flagship, E1-1…E1-9 with auto-filled energy/Scope figures + guided narrative editor), **VSME** Basic and Comprehensive modules (EFRAG SME standard), CRREM-aligned stranding, GRESB readiness, EnEfG measures plan, residential summary.
- **Authoring & output:** a structured **narrative editor** with PostgreSQL persistence (auto numbers
  - human narrative, no AI auto-draft), **per-building scoping**, and DOCX/PDF export.

3.6 IoT / fault detection

A real-time layer implementing **ASHRAE Guideline 36** automated fault detection (12 diagnostic rules, ~31 sensor types) with multi-protocol adapters for **BACnet / Modbus / MQTT**.

3.7 Residential vertical (Hausverwaltung)

A full second vertical for residential property management: unit-level data model, **heat-cost & CO_2-cost allocation**, a **landlord investment case** that resolves the split-incentive problem (who pays for retrofits vs. who benefits), and **climate-adjusted EUI** consistent with the commercial engine — with separate resident and manager views.

3.8 Execution marketplace (Phase 0)

The first step of the **insight → action → execution** loop: a lean, founder-brokered "Request an installer" capture on every recommendation, with honest framing ("EnergyLens introduces; it does not warrant or contract"). The roadmap — vendor matching, RFQ/quotes, take-rate commission — is a future revenue line (see §6).

3.9 Self-serve onboarding + Data Score

A gamified **Data Score** maps data completeness to report readiness: each building shows what is ready / partial / locked and exactly which input unlocks which report — an indirect incentive to improve data quality, and the foundation for fully automated, self-serve onboarding.

Maturity, stated plainly. The platform runs on **ten representative buildings** (Germany, Turkey, Austria, the Netherlands; ~3.5 years; ~693,000 data points). The **methodology and external data are real**; building-level operational data is **synthetic** until the first pilot. There are **no paying customers yet** — a first pilot is in active outreach. The funded year converts this proven prototype into a validated, incorporated company.

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4. Innovation & unique selling proposition

EnergyLens's innovation is not a single feature but a **combination no competitor offers at the mid-market**, backed by a research-grade development agenda. What already works and what the coming year develops are kept distinct below.

4.1 The unique combination (already working)

1. **Fabric-native, with auditable carbon lineage.** Because every workload reads one copy of the data on OneLake, every disclosed kilogram of CO_2 traces back to the raw bill row — the "regulator-grade lineage" ESRS/VSME reporting requires, and structurally hard for spreadsheet or multi-tool competitors to match.
2. **An AI copilot that uses tool use, not text summaries** — it runs real Lakehouse queries and one tool writes back, so it can act.
3. **Multi-protocol IoT fault detection at mid-market price** — ASHRAE Guideline 36 across BACnet, Modbus and MQTT, as affordable SaaS rather than a six-figure enterprise install.
4. **A multi-country EU-regulation & reporting engine** (DE, AT, NL, TR, EU) that turns findings into **ranked actions and the actual regulatory documents**.
5. **A closed insight → action → execution loop** — from anomaly to recommendation to a brokered installer, with a Data-Score onboarding that makes the whole thing self-serve.

4.2 The innovation agenda (developed during the funded year)

The year turns the working platform into a defensible, research-grade product along **four axes**:

- **A — Agentic AI copilot.** Evolve from answering questions to **proposing and sequencing** retrofit and dispatch plans — an autonomous "decarbonisation advisor" grounded in the building's own data.
- **P — Predictive intelligence.** Move beyond statistical forecasting to **ML predictive maintenance** that detects developing faults before they cost money.
- **O — Closed-loop optimisation.** Move from simulation to control: optimal battery dispatch, HVAC setpoint optimisation and **automated demand response**, aligned with the EU Demand-Response Network Code.
- **M — Auditable carbon methodology (academic IP).** Formalise the traceable carbon-lineage method into a **publishable methodology, developed jointly with the academic host** — strengthening both the product's defensibility and its research-transfer character.

4.3 Defensibility (moat)

The moat is the compounding combination: domain-specific data models + an auditable methodology (potential academic IP) + a Fabric-native architecture + an agentic AI layer + an EU-regulatory knowledge base + a two-sided execution loop, all aimed at a segment incumbents under-serve. None alone is unassailable; together, with first-mover reference customers in DACH, they create a defensible position. Microsoft-ecosystem alignment (DP-600, Partner Network track) adds distribution.

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5. Market & business model

5.1 Market size

- **External anchor:** the BEMS software market is independently estimated at **~US $8.4 bn (2026) → ~$24 bn (2035), ~8.7 % CAGR** (market-research consensus; figures vary by definition).
- **Illustrative top-down TAM/SAM:** ~5 million EU non-residential buildings at €100–700/building/month imply an order-of-magnitude **TAM ~€18 bn/yr**, with a DACH-focused **SAM ~€4.3 bn/yr**. These are illustrative framing figures, not bottom-up forecasts.
- **Serviceable obtainable (3-year, realistic):** 100–200 customers → €1–2 m ARR.

5.2 Customers & segments

Validation-first: start where deals close in 2–8 weeks — **university campuses** and **DACH mid-market property managers (5–50 buildings)** — then use case studies to reach hotels, healthcare and REITs. A distinct second vertical, **residential property management (Hausverwaltung)**, is already built and addresses a large, compliance-driven (CO2KostAufG) market. Enterprise procurement is deliberately deferred until product and references are mature.

5.3 Business model & pricing (per building / month)

Tier	Price	Target segment
Insight	€99	SME offices
Monitor	€299	mid-market property management
Copilot	€699+	enterprise, healthcare, IoT-equipped
Portfolio (custom)	€5–50k/mo	10+ building portfolios

A pilot is **free for the first three months** to win reference cases. A future **marketplace take-rate** (§6) adds a second, usage-aligned revenue line on top of subscriptions.

5.4 Unit economics (targets, pre-revenue)

Targets to validate in the first pilots, not results: ARR per customer ≈ **€17k** (5 buildings × €299/mo); gross margin **~85 %** (SaaS-typical minus cloud/inference); target **CAC payback < 6 months** in a founder-led, content- and network-driven motion; designed to break even on infrastructure cost around the second customer. Headline LTV/CAC multiples are deliberately omitted until real acquisition data exists.

5.5 Competitive landscape

Competitor	Position	Gap EnergyLens fills
Measurabl / Aquicore	US enterprise, well-funded	mid-market EU coverage
Cortexa	EU enterprise	mid-market accessibility & price
Schneider EcoStruxure	peer (reporting + AI)	multi-protocol IoT + Fabric-native + mid-market price
Siemens Desigo	BMS / building automation	the analytics/intelligence layer, not a BMS

One-line position: *the Microsoft Fabric-native energy-intelligence and EU-compliance platform for the European mid-market, with an AI copilot that reasons over the Lakehouse via tool use — from under €299/building/month.*

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6. Product roadmap & future development

The technical foundation is broad and already built; the roadmap is about **depth, validation and new revenue lines**, not rebuilding.

- **Execution marketplace (revenue line).** Evolve Phase 0 “request an installer” into a two-sided marketplace: vendor registry + matching, RFQ/quotes, and a **take-rate commission** on delivered retrofits — monetising the insight→action→execution loop beyond subscriptions.
- **Reporting engine.** Extend the 13-report engine toward **machine-readable filing (XBRL)**, additional standards and native Word/Excel output; deepen ESRS/VSME conformance as more client data arrives.
- **Residential expansion.** Grow the Hausverwaltung vertical: portfolio heat-cost automation, tenant-facing transparency, subsidy/retrofit ROI at scale.
- **Self-serve, fully automated onboarding.** Drive the Data-Score onboarding to a fully automated pipeline (customer data → reports with no founder in the loop) — a scale-economics step.
- **Innovation axes (A/P/O/M, §4.2).** Agentic planning copilot, ML predictive maintenance, closed-loop optimisation + demand response, and the published carbon methodology.
- **IoT depth.** BACnet + Modbus first (90 % of DACH BMS), then MQTT streaming and (premium) OPC-UA.
- **Geographic expansion.** Beyond DACH to BeNeLux + France (Décret Tertiaire, Erkende Maatregelen), with localisation.
- **Microsoft Partner Network.** Co-marketing and Cloud Marketplace distribution.

Long-horizon vision: by 2028, the leading Fabric-native energy-intelligence platform for European real estate — turning every building into a self-diagnosing, decarbonising asset.

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7. 12-month implementation plan

The technical foundation exists and the founder builds quickly; the genuine 12-month work is **customer validation, sales, incorporation, regulatory/IP and research**. Milestones are therefore expressed as business outcomes, with technical work in support. Because execution is fast, the plan is front-loaded and milestones can be reached early.

Q1 (months 1–3) — Validate. First pilot live; 2–3 design partners engaged; pricing in test. Build: production hardening; predictive-maintenance models in training; the carbon-methodology research started with the academic host. *Metric:* ≥ 1 live pilot, ≥ 5 discovery conversations/week.

Q2 (months 4–6) — Sell. Convert pilots into 2–3 paying customers; pricing validated. Build: ship the agentic copilot (propose-and-sequence) as the commercial wedge; harden the compliance engine for real client data. *Metric:* first recurring revenue; ≥ 1 named case study.

Q3 (months 7–9) — Incorporate & build the team. Register the **UG (haftungsbeschränkt)**; first hire (GTM / sales-engineering); pipeline toward 5 customers; launch marketplace matching as a pilot. Build: closed-loop optimisation in production; security/GDPR review. *Metric:* company registered; 4–5 customers signed or in late stage.

Q4 (months 10–12) — Fundable. 5+ customers, €5–10k MRR; the **auditable-carbon methodology paper submitted** with the academic host; Microsoft Partner Network track entered. Build: scale-readiness (automated onboarding, reliability); seed/angel materials. *Metric:* sustainable MRR; a defensible, demonstrable innovation; a credible seed story.

Traction before the year begins. First-pilot outreach, early sales conversations and freelance income proceed in parallel now, so the venture enters the funded period **with traction already in hand**.

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8. Use of funds (≈ €45,000 over 12 months)

The 12-month budget is modest and focused, split between the founder's full-time commitment, a small material budget and external coaching:

Category	Amount	Purpose
Stipend (living costs)	€30,000	12 months full-time (rent, living, insurance)
Software & cloud	€4,000	Microsoft Fabric capacity (F-SKU), Power BI, AI inference, SaaS tooling
Hardware	€2,500	Development laptop refresh, monitor, A/V for demos
Customer acquisition	€1,500	Targeted outreach, events, content
Legal & administrative	€2,000	UG incorporation, IP counsel, GDPR review
Coaching	€5,000	Start-up mentor, sales coaching, technical advisory
Total	**≈ €45,000**	toward an incorporated, revenue-generating company

A founding team would expand the material budget and add a stipend per member; bringing on a complementary commercial co-founder is an explicit option for the funded year (§9).

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9. Team, risks & mitigations

9.1 Team & hiring plan

The venture is currently led by a **single founder**. This is mitigated deliberately: the founder already spans the two hardest-to-combine skill sets (**energy domain + data engineering**), reducing early dependency on hires; an academic mentor and 1–2 advisors provide a visible support structure; a **first GTM / sales-engineering hire is planned for Q3**, timed to first revenue; and a complementary commercial **co-founder is an open option** once early traction de-risks equity.

9.2 Risks & mitigations

Risk	Mitigation
Solo-founder bandwidth	First GTM hire in Q3; advisor network; onboarding automation
Pilot → paid conversion is slow	Free, low-risk pilots; quantified value reports; freelance runway buffer
"Integration, not innovation" perception	Four-axis agenda (agentic AI, predictive, optimisation, methodology IP) + an academic paper
Methodology / compliance over-claim	Honest labelling ("ESRS-E1-aligned support"; screening estimates flagged) reviewed before any "compliant" wording
Visa / residency	18-month post-graduation Job-Search Visa; intent to incorporate (UG) and remain in Germany
Single-vendor (Microsoft) dependency	Provider-abstracted AI layer; standards-based IoT; data portable via open Delta/Parquet
Incumbent price pressure	EU-regulatory depth + Fabric-native lineage + mid-market focus as differentiation, not price

9.3 Current limitations (stated openly)

A credible application is precise about maturity: Scope 3 is a labelled **screening estimate** (not disclosure-grade); market-based Scope 2 needs supplier-contract data to differ from location-based; **CRREM pathways are indicative** pending a License-Partner agreement; some HVAC splits are **modelled** (HDD/CDD) rather than sub-metered; battery-ROI savings are currently **upper-bound** pending hour-by-hour tariff matching; building consumption data is **synthetic** until the first real pilot. Each has a concrete improvement on the roadmap. Naming exactly what is not yet real, and exactly how to make it real, is a stronger signal than a polished claim of completeness.

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10. Founder & academic anchor

Ali Mert Özdemir.

- **Education:** B.Sc. Energy Engineering (Yaşar University, İzmir); M.Sc. Energy Management (BSBI Berlin, June 2026).
- **Certification: Microsoft DP-600** (Implementing Analytics Solutions Using Microsoft Fabric) — a recent, in-demand certification rarely held by solo founders.

- **Demonstrated execution:** designed and built the **entire EnergyLens stack** — Fabric pipeline, 10-page Power BI report, IoT/fault-detection layer, battery simulator, multi-tenant web app, AI copilot and a 13-report compliance engine — solo, end-to-end, accelerated by AI-assisted development.
- **Residency:** Berlin-based; 18-month Job-Search Visa runway post-graduation to incorporate and reach first revenue in Germany.

Academic anchor. EnergyLens originates directly from the founder's M.Sc. work, which gives the venture a genuine **research-transfer character**: a **joint carbon-methodology publication** with the academic host is planned during the funded year (innovation axis M, §4.2). *[Academic host and named mentor — to be confirmed.]*

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11. Timeline & next steps

- **Now:** confirm the academic host and named mentor; finalise this package with the host's start-up service.
- **In parallel:** first pilot, early customer outreach and freelance runway — building traction before the funded period begins.
- **On confirmation:** submit through the host institution; on approval, sign the funding agreement and begin the 12-month period (target Q1 2027).

EnergyLens — turning every building into a self-diagnosing, decarbonising asset. Figures marked illustrative or target are pre-revenue assumptions to be validated in pilots. energylens.eu · Microsoft DP-600.